

H8 Dataset Creation

Team BRSM Team BNT

```
library(dplyr)

##
## Attaching package: 'dplyr'
##
## The following objects are masked from 'package:stats':
##
##   filter, lag
##
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(readr)
library(stringr)

# Bhavani - Dataset creation
process_file <- function(file_path) {

  # reading the log files and finding the blocks
  df <- read_csv(file_path, show_col_types = FALSE)
  colnames(df) <- make.names(colnames(df))
  practice_index <- max(which(df$Event %in% c(
    "Practice Sentence shown",
    "Practice IR pressed",
    "Practice WR pressed",
    "Practice Missed"
  )))

  rest_indices <- which(df$Event == "Rest Phase started")
  block1_range <- (practice_index + 1):(rest_indices[1] - 1)
  block2_range <- (rest_indices[1] + 1):(rest_indices[2] - 1)
  block3_range <- (rest_indices[2] + 1):nrow(df)
  block1 <- df[block1_range, ] %>% mutate(Block_Number = 1)
  block2 <- df[block2_range, ] %>% mutate(Block_Number = 2)
  block3 <- df[block3_range, ] %>% mutate(Block_Number = 3)

  blocks <- list(block1, block2, block3)

  valid_blocks <- list()

  # validating each block using the formula in the given txt file
  for (block in blocks) {
    target_trials <- block %>% filter(isTarget == TRUE, Event == "Sentence shown")
    if (nrow(target_trials) != 32) {
      next
    }
  }
}
```

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}

correct_val <- sum((block$isValidation == TRUE & block$Event == "Validation IR pressed"), na.rm = TRUE)
wrong_val <- sum((block$Event == "Validation Wrong IR pressed"), na.rm = TRUE)
missed_val <- sum((block$isValidation == TRUE & block$Event == "Validation Missed"), na.rm = TRUE)
if (correct_val > ((wrong_val / 2) + missed_val)) {
  valid_blocks <- append(valid_blocks, list(block))
}
}

# combining the blocks (stacking rows), removing rows that are for gap time, and keeping only the target
if (length(valid_blocks) == 0){
  return(NULL)
}
valid_data <- bind_rows(valid_blocks)
valid_data <- valid_data %>% filter(isTarget == TRUE, Event != "gap_time")

# Removing missing values and outliers
valid_data <- valid_data %>% filter(participant_ID != "N/A", Timestamp != "N/A")
valid_data <- valid_data %>%
mutate(
  Reaction_time_IR = na_if(Reaction_time_IR, "N/A"),
  Reaction_time_WR = na_if(Reaction_time_WR, "N/A"),
  Reaction_time_IR = as.numeric(Reaction_time_IR),
  Reaction_time_WR = as.numeric(Reaction_time_WR)
)
valid_data <- valid_data %>%
filter(
  (is.na(Reaction_time_IR) | (Reaction_time_IR >= 150 & Reaction_time_IR <= 3000)) &
  (is.na(Reaction_time_WR) | (Reaction_time_WR >= 150 & Reaction_time_WR <= 3000))
)
valid_data <- valid_data %>%
filter(Button=="Yes" | Button=="No" | Button=="None" | Button=="N/A" | Button=="Spacebar")

return(valid_data)
}

log_files <- list.files("Sentence Memorability/NewLogsAnonymized", pattern = "\\..log$", full.names = TRUE)

# applying the block creation and preprocessing on all the log files
all_data <- log_files %>% lapply(process_file) %>% bind_rows()
write_csv(all_data, "sentence_memorability_data.csv")

```